collateral is worth $1,500. Now, the investor can increase the size of their loan. To maintain the collateralization of 200 percent, the investor can mint an extra 250 DAI.

A more interesting scenario is when the value of the collateral drops. Suppose the value of the ETH drops by 25 percent, from $200 to $150. In this case, the value of the collateral drops to $750 and the collateralization ratio drops to 1.5 ($750/1.5 = 500).

The vault holder faces three scenarios. First, they can increase the amount of collateral in the contract (by, e.g., adding 1 ETH). Second, they can use the 500 DAI to pay back the loan and repatriate the 5 ETH. These ETH are now worth $250 less, but the depreciation in value would have happened regardless of the loan. Third, the loan is liquidated by a *keeper* (any external actor) who is incentivized to find contracts eligible for liquidation. The keeper auctions the ETH for enough DAI to pay off the loan. In this case, 3.33 ETH would be sold and 1.47 would be returned to the vault holder (the keeper earns an incentive fee of 0.2 ETH). The vault holder then has 500 DAI worth $500 and 1.47 ETH worth $220. This analysis does not include gas fees.

Two forces in this process reinforce the stability of DAI: overcollateralization and market actions. In the liquidation, ETH are sold and DAI are purchased, which exerts positive price pressure on DAI. This simple example does not address many features in the MakerDAO ecosystem (Figure 6.1), in particular the fee mechanisms and the debt limit, which we will now explore.